

Instructions

Entity Relationship Diagram (data modeler)(MLD)

– Write the sentences for each relationship

Each Preference may belong to one and only one member. Each Member may enter one or more Preference.

Each Member may can request one or more Reservation. Each Reservation must be made by one and only one Member.

Each Member may buy one or more Purchase. Each Purchase must be record by one and only one Member.

Each Member may place one or more Book Order. Each Book Order must be placed by one and only one Member.

Each Member may have it's record maintained by one or more Librarian. Each Librarian may manage by one and only one Member.

Each Member may create one or more borrowing. Each borrowing must belong to one and only one Member.

Each Book may can have one or more Reservation. Each Reservation must be recorded by one and only one Book.

Each Book may can be purchased by one or more Purchase. Each Purchase must be recorded by one and only one Book.

Each Book may can be ordered by one or more Book Order. Each Book Order must be for one and only one Book.

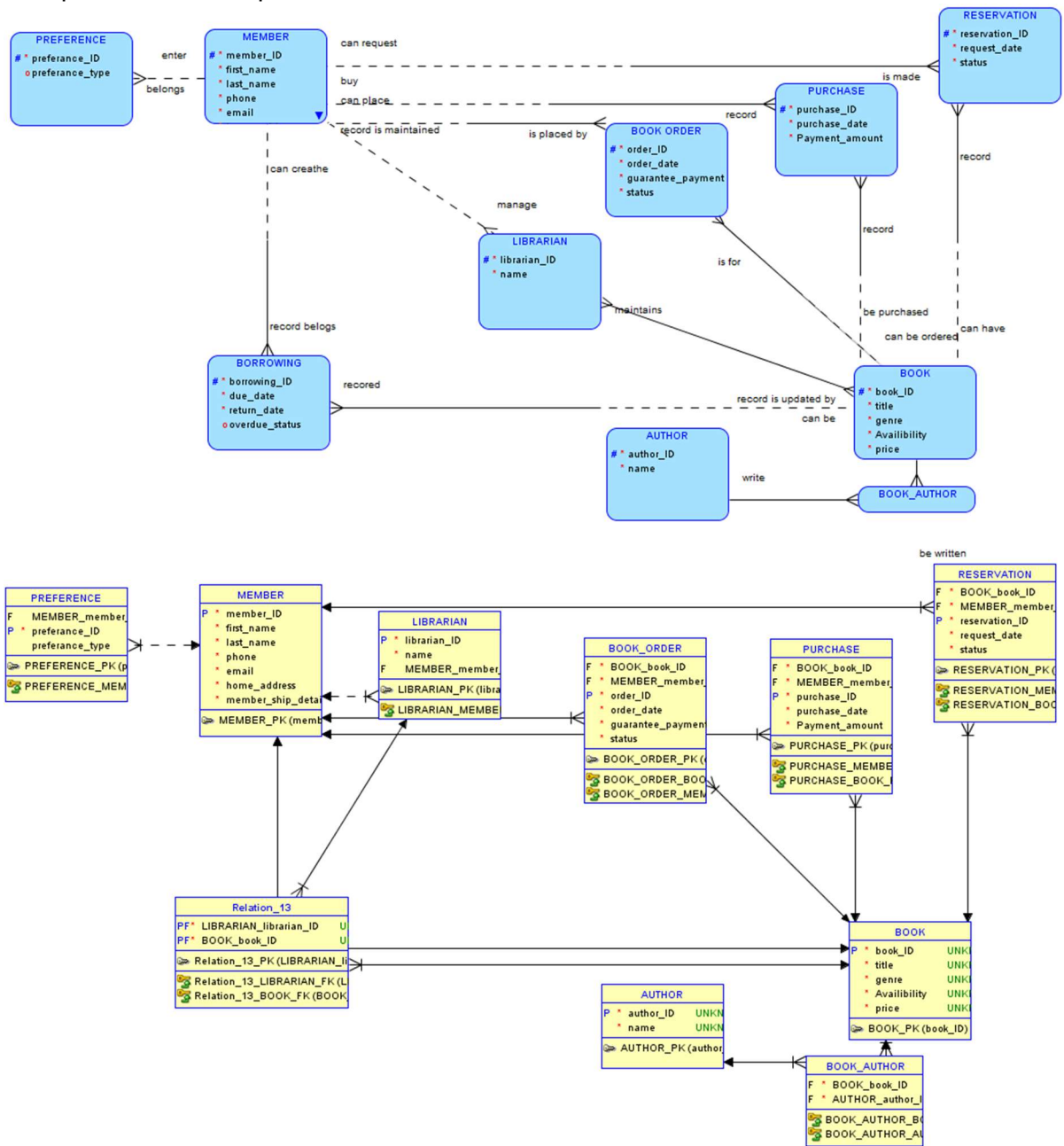
Each Book may can have it's record updated by one or more borrowing. Each borrowing must be recorded for one and only one Book.

Each Book must have it's record updated by one or more Librarian. Each Librarian must maintain one or more Book.

Each Book must be written by one or more Author. Each Author must write one or more Book.

Laura Glode PROJECT

– Map the relationships.



– Draw the matrix diagram

	Member	Preference	Reservation	Book Order	borrowing	Librarian	Author	Book	Purchase
Member		enter	request	place	create	record			buy
Preference	belong								
Reservation	made							recorded	
Book Order	placed							must be	
borrowing	belong							recorded	
Librarian	manage								maintain
Author								write	
Book			have	ordered	record update	record updated	written		purchased
Purchase	recorded							recorded	

MAPPING RELATIONSHIPS					
Preference			Book Order		
Key-Types	Optimality	Column-name	Key-Types	Optimality	Column-name
PK	#	Preference_ID	PK	#	Book_Order_ID
	o	Preference_type	FK	#	Member_ID
FK	#	Member_ID	FK	#	Book_ID
				*	order_date
				*	garantee_payment
				*	home_address
				*	member_ship_data
Member			Purchase		
Key-Types	Optimality	Column-name	Key-Types	Optimality	Column-name
PK	#	Member_ID	FK	#	Book_Order_ID
	*	first_name	FK	#	Member_ID
	*	last_name	PK	#	Purchase_ID
	*	phone		*	Purchase_date
	*	email		*	payment_amout
	*	home_address			
	*	member_ship_data			
Librarian			Reservation		
Key-Types	Optimality	Column-name	Key-Types	Optimality	Column-name
PK(1)	#	Librarian_ID	FK	#	Book_Order_ID
PK(12)	#	Member_ID	FK	#	Member_ID
	*		PK	#	Reservation_ID
				*	request_date
				*	status
			FK	#	Author_ID
Book			Author		
Key-Types	Optimality	Column-name	Key-Types	Optimality	Column-name
PK(1)	#	Book_ID	PK(1)	#	Author_ID
PK(12)	#	title	PK(12)	#	Book_id
	*	genre			
		Availiblity			
		Price			name